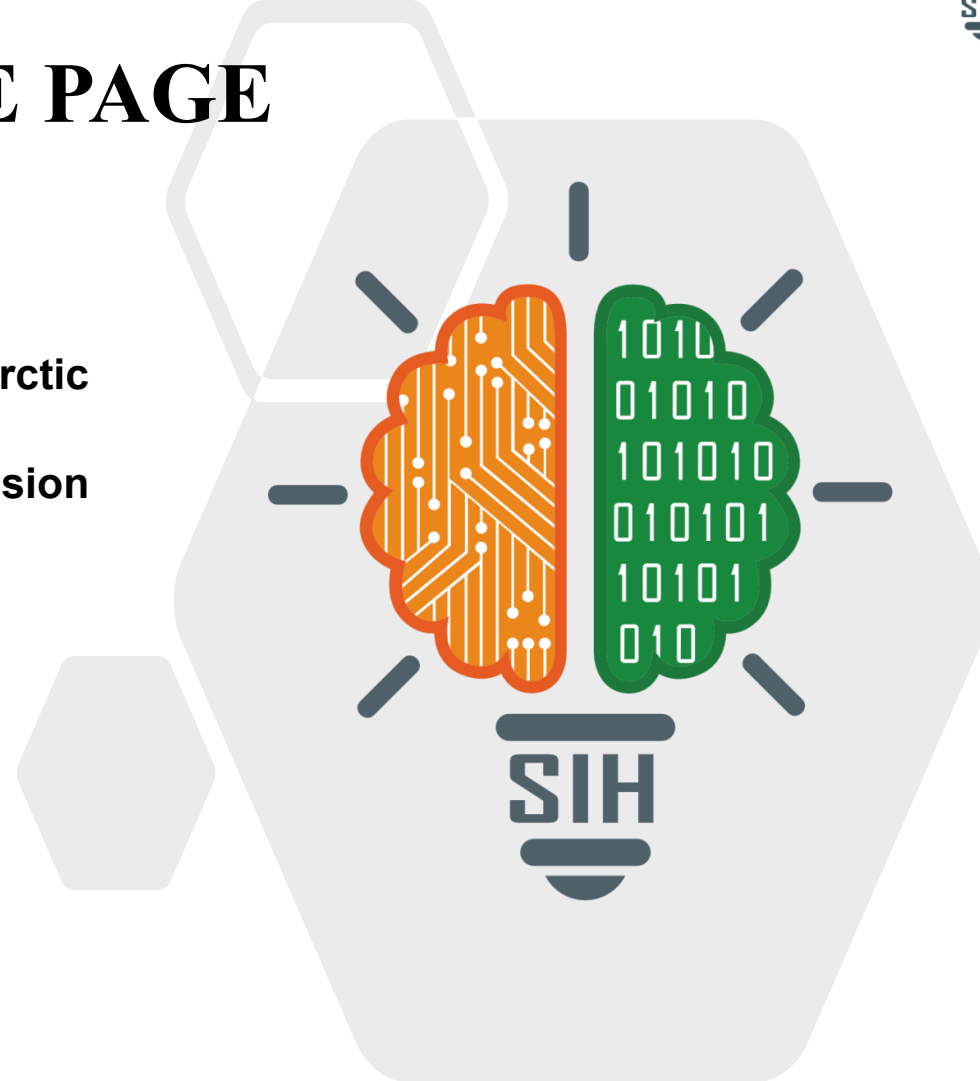


TITLE PAGE

- **Problem Statement ID:** SIH26059
- **Problem Statement Title:** AI-Enabled Antarctic Sea-Ice, Iceberg Trajectory, and Navigation Decision Support System.
- **Theme:** Transportation & Logistics
- **PS Category:** Software
- **Team ID:**
- **Team Name:** Odyssey Prime



❖ Proposed Solution

- **Integrated Intelligence:** Antarctic Traverse integrates satellite, sea-ice, ocean, and weather data to forecast ice conditions, detect icebergs, and predict their movement.
- **From Prediction to Action:** These forecasts are transformed into a dynamic hazard and risk map to enable safer navigation for Antarctic research vessels.
- **Unified Decision Support:** AI-based forecasting, physics-informed iceberg trajectory modelling, and explainable, risk-aware routing are integrated into a unified GIS-based decision-support workflow.

Outcome: One Unified view of hazards, route risks and safer alternatives.

- **Technology Stack**

Python • PyTorch / TensorFlow • XGBoost • GeoPandas / Rasterio • QGIS/Leaflet • FastAPI

- **Core Models & Algorithms**

U-Net • ConvLSTM • Optical Flow • Physics-informed ML with residual correction • A*/Dijkstra • NSGA-II

Data Sources

Sentinel-1
NSIDC/AMSR2
ERA5 • GLORYS

Prediction

Sea-Ice Concentration
Ice Berg
Trajectory Forecasting

Risk Assessment

Risk Index
Hazard Score
Uncertainty

Route Optimization

Dynamic Graph
A*/Dijkstra
NSGA-II

Decision Dashboard

GIS-based Map
Safe Route
Alternatives

Method: Ingest → Align → Predict → Quantify Risk → Optimize → Visualize

- **Feasibility:** Public datasets and reproducible models make a practical, modular MVP feasible.
- **Key Risks:** Sparse Antarctic ground truth, ice-edge bias, satellite data latency, and computational constraints.
- **Mitigation:** Phase the development: start with U-Net/ConvLSTM, then add uncertainty-aware outputs and physics-informed drift modelling; validate against BYU/NIC, SCAR, and labelled Sentinel-1 data.

- **Safer Antarctic Voyages:** Forecasted sea-ice conditions, iceberg detection, and trajectory prediction enable safer route decisions for research vessels.
- **Better Operational Decisions:** A unified GIS dashboard brings forecast layers, hazard/risk scores, confidence and ranked route alternatives into one operator-facing view.
- **Fuel & Environmental Efficiency:** Multi-objective routing balances risk, travel time, and fuel consumption, enabling operators to compare safer, more efficient route alternatives.

1. **Braakmann-Folgmann et al. (2023):** *Mapping the extent of giant Antarctic icebergs with deep learning*
2. **Andersson et al. (2021):** *Seasonal Arctic sea ice forecasting with probabilistic deep learning*
3. **Wang et al. (2023):** *Subseasonal Prediction of Regional Antarctic Sea Ice by a Deep Learning Model*
4. **Hui et al. (2017):** *Satellite-Based Sea Ice Navigation for Prydz Bay, East Antarctica*
5. **Liu et al. (2025):** *Vessel Safety Navigation Under the Influence of Antarctic Sea Ice*
6. **Budge & Long (2018):** *A Comprehensive Database for Antarctic Iceberg Tracking Using Scatterometer Data*